pharmaniaga®

Leucovorin Calcium 50mg/5mL Injection

COMPOSITION
Each 5mL contains Leucovorin Calcium equivalent to Leucovorin 50.00mg and also contains preservatives i.e. Methyl Paraben 4.00mg and Propyl Paraben 1.00mg.

PRODUCT DESCRIPTION
A clear, pale yellow coloured solution.

PHARMACODYNAMICS
Leucovorin calcium is the calcium salt of 5-formyl tetrahydrofolic acid. It is an active metabolite of folinic acid and an essential coenzyme for nucleic acid synthesis in cytotoxic therapy.
Leucovorin calcium is frequently used to diminish the toxicity and counteract the action of folate antagonists, such as methotrexate. Leucovorin calcium and folate antagonists share the same membrane transport carrier and compete for transport into cells, stimulating folate antagonist efflux. It also protects cells from the effects of folate antagonist by repletion of the reduce folate pool. Leucovorin calcium serves as a pre-reduced source of H4 folate; it can therefore bypass folate antagonist blockage and provide a source for the various coenzyme forms of folic acid.

Leucovorin calcium is also frequently used in the biochemical modulation of fluoropyridine (5-FU) to enhance its cytotoxic activity. 5-FU inhibits thymidylate synthase (TS), a key enzyme involved in pyrimidine biosynthesis, and leucovorin calcium enhances TS inhibition by increasing the intracellular folate pool, thus stabilising the 5FU-TS complex and increasing activity.

Finally intravenous leucovorin calcium can be administered for the prevention and treatment of folate deficiency when it cannot be prevented or corrected by the administration of folic acid by the oral route. This may be the case during total parenteral nutrition and severe malabsorption disorders. It is also indicated for the treatment of megaloblastic anaemia due to folic acid deficiency, when oral administration is not feasible.

PHARMACOKINETICS
Absorption
Following intramuscular administration of the aqueous solution, systemic availability is comparable to an intravenous administration. However, lower peak serum levels (Cmax) are achieved.

Metabolism
Leucovorin calcium is a racemate where the L-form (L-5-formyl-tetrahydrofolate, L-5-formyl-THF), is the active enantiomer. The major metabolic product of folinic acid is 5-methyl-tetrahydrofolic acid (5-methyl-THF) which is predominantly produced in the liver and intestinal mucosa.

Distribution
The distribution volume of folinic acid is not known.
Peak serum levels of the parent substance (D/L-5-formyl-tetrahydrofolic acid, folinic acid) are reached 10 minutes after i.v. administration.
AUC for L-5-formyl-THF and 5-methyl-THF were 28.4±3.5 mg.min/l and 129±112 mg.min/l after a dose of 25 mg. The inactive D-isomer is present in higher concentration than L-5-formyltetrahydrofolate.

Elimination
The elimination half-life is 32 - 35 minutes for the active L-form and 352 - 485 minutes for the inactive D-form, respectively.
The total terminal half-life of the active metabolites is about 6 hours (after intravenous and intramuscular administration).

Excretion
80-90 % with the urine (5- and 10-formyl-tetrahydrofolates inactive metabolites), 5-8 % with the faeces.

INDICATION
- To diminish the toxicity and counteract the effects of inadvertently administered over dosage of folic acid antagonists.
- As a rescue after high dose methotrexate therapy.
- In the treatment of megaloblastic anaemia due to sprue, nutritional deficiency, pregnancy and infancy when oral therapy is not possible.
- For use in combination with fluorouracil to prolong survival in the palliative treatment of patients with advanced colorectal cancer.

RECOMMENDED DOSAGE
<u>Impaired Methotrexate Elimination or Inadvertent Overdosage:</u>
Leucovorin calcium rescue should begin as soon as possible after an inadvertent overdosage and within 24 hours of methotrexate administration when there is delayed excretion. Leucovorin calcium 10 mg/m^2 should be administered IV or IM every 6 hours until the serum methotrexate level is less than 10^{-8}M.

Serum creatinine and methotrexate levels should be determined at 24 hour intervals. If the 24 hours serum creatinine has increased 50% over baseline or if the 24 hour methotrexate level is greater than 5 x 10^{-6}M or the 48 hour level is greater than 9 x 10^{-7}M, the dose of Leucovorin calcium should be increased to 100 mg/m^2 IV every 3 hours until the methotrexate level is less than 10^{-8}M.
Hydration (3L/d) and urinary alkalinisation with sodium bicarbonate solution should be employed concomitantly. The bicarbonate dose should be adjusted to maintain the urine pH at 7.0 or greater.

<u>Leucovorin Rescue After High-dose Methotrexate Therapy:</u>
The recommendations for Leucovorin rescue are based on a methotrexate dose of 12 to 15 gm/m^2 administered by intravenous infusion over 4 hours. Leucovorin rescue at a dose of 15mg (approximately 10 mg/m2) every 6 hours for 10 doses starts 24 hours after the beginning of the Methotrexate infusion. Serum creatinine & methotrexate levels should be determined at least once daily. Leucovorin administration, hydration, and urinary alkalinisation (pH of 7.0 or greater) should be continued until the methotrexate level is below 5 x 10^{-8} M (0.05 micromolar).

<u>Megaloblastic Anaemia due to Folic Acid Deficiency:</u>
Upto 1 mg daily. There is no evidence that doses greater than 1 mg/day have greater efficacy than those of 1 mg, additionally loss of folate in urine becomes roughly logarithmic as the amount administered exceeds 1 mg.

<u>Advanced Colorectal Cancer:</u>
Either of the following two regimens is recommended:
1. Leucovorin is administered at 200 mg/m^2 by slow intravenous injection over a minimum 3 minutes, followed by 5-fluorouracil at 370 mg/m^2 by intravenous injection.
2. Leucovorin is administered at 20 mg/m^2 by intravenous injection followed by 5-fluorouracil at 425 mg/m^2 by intravenous injection.

   Treatment is repeated daily for 5 days. This 5-treatment course may be repeated at 4-week (28 days) intervals for two courses and then repeated at 4 to 5 week (28-35 days) intervals provided that the patient has completely recovered from toxic effects of the prior treatment course.

In subsequent treatment courses, the dosage of 5-fluorouracil should be adjusted based on patient intolerance of the prior treatment course. The daily dosage of 5-fluorouracil should be reduced by 20% for patients who experienced moderate haematologic or gastrointestinal toxicity in the prior treatment course and by 30% for patients who experienced severe toxicity. For patients who experienced no toxicity in prior treatment course, 5-fluorouracil dosage may be increased by 10%. Leucovorin dosage are not adjusted for toxicity.

ROUTE OF ADMINISTRATION
Parenteral

CONTRAINDICATIONS
1. It is contraindicated in patients who have known hypersensitivity to Leucovorin.
2. Megaloblastic anaemia due to Vitamin B12 deficiency.
3. Pernicious anaemia due to Vitamin B12 deficiency.

WARNINGS AND PRECAUTIONS
Leucovorin calcium is not suitable for the treatment of pernicious anaemias and other anaemias resulting from lack of Vitamin B12. Haematological remissions may occur, while the neurological manifestations remain progressive. Simultaneous therapy with a folic acid antagonist and leucovorin calcium is not recommended because the effect of the folic acid antagonist is either reduced or completely inhibited.

Because of the Ca^{2+} content of the leucovorin calcium injections, no more than 16 mL of the 10 mg/mL formulations (160 mg of leucovorin calcium) should be injected intravenously per minute.

Leucovorin calcium may enhance the toxicity of fluorouracil. Deaths from severe enterocolitis, diarrhoea and dehydration have been reported in elderly patients receiving leucovorin calcium and fluorouracil.
Concomitant granulocytopenia and fever were present in some, but not all, of the patients.
Seizures and/or syncope have been reported rarely in cancer patients receiving leucovorin calcium, usually in association with fluoropyrimidine administration, and most commonly in those with CNS metastases. Since three patients had recurrent neurological symptoms on rechallenge with leucovorin calcium, further treatment with leucovorin calcium is not recommended in these circumstances.

Parenteral administration is preferable to oral dosing if there is a possibility that the patient may vomit or not absorb the leucovorin calcium.

Leucovorin calcium has no effect on non-haematological toxicities of methotrexate, such as the nephrotoxicity resulting from drug and/or metabolite precipitation in the kidney.

Leucovorin calcium should only be used with folic acid antagonists, e.g. methotrexate, or fluoropyrimidines, e.g. 5-fluorouracil, under the direct supervision of a clinician experienced in the use of cancer chemotherapeutic agents.

Combination therapy with 5-fluorouracil:
Leucovorin calcium enhances the toxicity of 5-fluorouracil. When these drugs are administered concurrently in the palliative therapy of advanced colorectal cancer, the dosage of 5-fluorouracil should be reduced accordingly.

Although the toxicities observed in patients treated with the combination of leucovorin followed by 5-fluorouracil are qualitatively similar to those observed in patients treated with 5-fluorouracil alone, gastrointestinal toxicities (particularly stomatitis and diarrhoea) are observed more commonly, may be more severe and of prolonged duration in patients treated with the combination.

Combination therapy with leucovorin calcium /5-fluorouracil must not be initiated or continued in patients who have symptoms of gastrointestinal toxicity of any severity, until those symptoms have completely resolved. Patients with diarrhoea must be monitored with particular care until the diarrhoea has resolved, as rapid clinical deterioration leading to death can occur.

Particular care should be taken when treating elderly or debilitated colorectal cancer patients with leucovorin calcium /5-fluorouracil, as these patients may be at increased risk of severe toxicity, particularly severe gastrointestinal toxicity.

INTERACTIONS WITH OTHER MEDICAMENTS

Folic acid in large amounts may counteract the antiepileptic effect of phenobarbital, phenytoin and primidone, and increase the frequency of seizures in susceptible children.

High oral, intravenous or intramuscular doses of leucovorin calcium may reduce the efficacy of intrathecally administered methotrexate.

Leucovorin calcium may enhance the toxicity of fluorouracil.

Incompatibilities

Leucovorin calcium has been reported to be incompatible with droperidol and phosphonosulphate.

PREGNANCY AND LACTATION

Pregnancy
Adequate animal reproduction studies have not been conducted with leucovorin. It is also not known whether leucovorin can cause foetal harm when administered to a pregnant woman or can affect reproduction capacity. Leucovorin should be given to a pregnant woman only if clearly needed.

Lactation
It is not known whether this drug is excreted in human milk, therefore, caution should be exercised when leucovorin is administered to a nursing mother.

SIDE EFFECTS

Occasional hypersensitivity, including anaphylactic reactions, has been reported; pyrexia has occurred rarely after injections. Gastrointestinal disturbances, insomnia, agitation and depression have been reported rarely, after high doses.

OVERDOSE AND TREATMENT

Excessive amounts of leucovorin may nullify the chemotherapeutic effect of folic acid antagonists.

Effect on ability to drive and use of machines

There is no evidence that leucovorin calcium has an effect on the ability to drive or use machines.

Instruction for Use and Handling

Prior to administration, leucovorin calcium should be inspected visually. The solution for injection or infusion should be a clear and yellowish solution. If cloudy in appearance or particles are observed, the solution should be discarded.

STORAGE CONDITIONS

Store in refrigerator (2°C – 8°C).
Protect from light. Do not freeze.
Single use only. Discard any unused portion.

SHELF LIFE

24 months

DOSAGE FORMS AND PACKAGING AVAILABLE:

- 10 x 5mL vial (amber).

PRODUCT REGISTRATION HOLDER /MANUFACTURER

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